UNIT 1 · TOPIC 1.2

Suspicious Website Logins

Quiz

Instructions: This quiz has 18 multiple-choice questions, 1 point each. Circle the letter of the single best answer for each question.

Name: **Date:** **Class:**

1. A student types a website address into their browser to log into their email. They notice the address bar shows "mail-gooogle-login.com" instead of the usual site. What is this an example of?

- ○ A. A normal website redirect for mobile users
- ○ B. A browser cache error
- ○ C. A fake login page used to harvest credentials
- ○ D. A VPN connection notice

2. Which of the following is the most reliable visual indicator that a login page may not be legitimate?

- ○ A. The page loads slowly
- ○ B. The URL does not match the official domain of the service
- ○ C. The page uses a dark color theme
- ○ D. The page asks for a username and password

3. What term describes the practice of an adversary setting up a fraudulent website that mimics a real one in order to steal usernames and passwords?

- ○ A. Load balancing
- ○ B. Credential harvesting
- ○ C. Two-factor authentication
- ○ D. Data encryption

4. A user reviews their account login history and finds entries from IP address 142.54.195.17 mixed in among entries from their usual IP address, 208.104.29.211, at times when they were in school and could not have logged in. What does this pattern most likely indicate?

- ○ A. The user logged in from a different device they forgot about
- ○ B. The internet service provider rotated the user's IP address automatically
- ○ C. The website is logging duplicate entries due to a software bug
- ○ D. The account credentials have likely been compromised and used by an unauthorized party

5. An adversary harvests a user's password from a fake login page and then successfully logs into three other unrelated services using the same credentials. Which weakness in user behavior did the adversary exploit?

- ○ A. Logging out after each session
- ○ B. Use of a strong, unique password
- ○ C. Enabling multi-factor authentication
- ○ D. Password reuse across multiple accounts

6. A user enters their credentials into a fake login page believing it is legitimate. Why does this technique succeed even against users who know phishing exists?

- ○ A. Because phishing only targets users with weak antivirus software
- ○ B. Because the fake page is designed to be visually and structurally indistinguishable from the real one under quick inspection
- ○ C. Because browsers cannot detect any unencrypted website
- ○ D. Because the user's password was already publicly known

7. A user's account login history shows consistent logins from one IP address on weekends and evenings, with two anomalous logins from a second IP address on a Wednesday and Monday, both during school hours. The user also reports that "some previous choices" in their account settings were changed without their knowledge. Which conclusion is best supported by the full body of evidence?

- ○ A. The anomalous logins are inconclusive on their own and the setting changes are unrelated, since settings can change due to app updates
- ○ B. The pattern of anomalous timing, a distinct IP address, and unauthorized setting changes together indicate the account was accessed and modified by someone other than the user
- ○ C. The second IP address indicates the user logged in from a school computer, which explains both the timing and the setting changes
- ○ D. The evidence only proves a login occurred at an unusual time; it cannot indicate who performed the login

8. A security analyst is asked to evaluate why credential-harvesting attacks remain effective despite widespread public awareness campaigns about phishing. Which explanation best reflects the underlying risk dynamic, rather than a surface-level cause?

- ○ A. Awareness campaigns fail because most users do not read them
- ○ B. Awareness reduces but does not eliminate risk, because attackers continuously adapt visual and contextual cues faster than users adapt their scrutiny, and a single lapse is sufficient for compromise
- ○ C. Credential harvesting is only effective against older or less tech-literate users
- ○ D. Awareness is irrelevant because all credential theft happens through malware, not phishing pages

9. Two users have identical passwords stolen via the same credential-harvesting campaign. User X used that password on only one account; User Y reused it across five linked accounts. Evaluate the difference in resulting risk impact for each user.

- ○ A. Neither user faces meaningful risk if their passwords were "strong" at the time of theft
- ○ B. Both users face identical risk, since the same password was stolen from both
- ○ C. User X faces greater risk because having only one account makes that account a more obvious target
- ○ D. User Y faces substantially greater impact because account linkage allows a single compromised credential to cascade into multiple account takeovers, while User X's exposure is contained to one account

10. Which of the following is a security control that strengthens authentication beyond just a password?

- ○ A. Multi-factor authentication (MFA)
- ○ B. Increasing the website's font size
- ○ C. Browsing in a private/incognito window
- ○ D. Clearing browser cookies daily

11. What tool helps users avoid the risky practice of reusing the same password across multiple accounts?

- ○ A. A file compression utility
- ○ B. A pop-up blocker

- ○ C. A screen recorder
- ○ D. A password manager

12. Before entering login credentials on a website, which security control can a user apply to verify the site's legitimacy?

- ○ A. Confirming the page loads in under one second
- ○ B. Confirming the page has a colorful logo
- ○ C. Checking that the URL exactly matches the official domain
- ○ D. Confirming ads are displayed on the page

13. A user's password was stolen via a fake login page, but the adversary was still unable to access the account because a one-time code from the user's phone was also required. Which control prevented account takeover in this case, and why?

- ○ A. Multi-factor authentication, because it requires a second proof of identity that the adversary did not also obtain
- ○ B. Password complexity requirements, because the original password was strong
- ○ C. HTTPS encryption, because it hides the password from the adversary
- ○ D. A firewall, because it blocked the adversary's connection attempt

14. After noticing unfamiliar logins in their account history, a user wants to apply controls to prevent future unauthorized access. Which combination of actions best addresses the root cause of the compromise rather than just the symptom?

- ○ A. Change the account's display name to make it harder for the adversary to find
- ○ B. Log out of the current session and wait to see if the unfamiliar logins stop on their own
- ○ C. Change the password to a new unique one and enable multi-factor authentication on the account
- ○ D. Delete the account activity log so the adversary cannot see when the user is online

15. A school requires students to use a password manager and enable MFA on their accounts. Explain why requiring both controls together is more effective than requiring either one alone.

- ○ A. Both controls do the same thing, so requiring both is redundant but harmless
- ○ B. A password manager reduces the chance of weak or reused passwords being harvested, while MFA limits the damage if a password is harvested anyway, the two controls address different points of failure
- ○ C. A password manager makes MFA unnecessary because it already generates secure passwords
- ○ D. MFA makes the password manager unnecessary because the second factor replaces the need for a password entirely

16. A platform offers two MFA options: SMS-based one-time codes and an authenticator app generating time-based codes. Evaluate which option provides stronger protection against credential-harvesting-based account takeover, and why.

- ○ A. Both options are equally vulnerable, so the choice between them makes no practical difference
- ○ B. SMS-based codes are always stronger because they require physical possession of the phone itself
- ○ C. The authenticator app is generally stronger, because SMS codes can be intercepted through SIM-swapping or number-porting attacks that don't require compromising the device itself
- ○ D. Neither option meaningfully improves security if the original password was already harvested

17. An organization is deciding between two policies: (1) mandatory password rotation every 30 days with no MFA, or (2) no forced rotation but mandatory MFA on all accounts. Evaluate which policy more effectively mitigates the risk demonstrated by credential-harvesting attacks.

- ○ A. Policy 1 is more effective, because frequent rotation guarantees that any harvested password will expire quickly
- ○ B. Policy 2 is more effective, because frequent password rotation does not stop a freshly harvested password from being used immediately, while MFA blocks unauthorized access regardless of when the password was captured
- ○ C. The two policies are equivalent in effectiveness against credential harvesting
- ○ D. Policy 1 is more effective because it requires no additional device or app for users to manage

18. A user wants to document the mitigation strategy they applied after discovering unauthorized logins on their account. Which documentation best captures both the immediate response and the layered prevention strategy, suitable for evaluating its effectiveness later?

- ○ A. "Told a friend about the suspicious activity"
- ○ B. "Logged out of the account"
- ○ C. "Changed password to a unique, manager-generated one; enabled app-based MFA; reviewed account activity log for further anomalies after the change"
- ○ D. "Waited one week to see if the activity continued"